

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Mock Interview

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 3 – Mock Interview
Weightage:	30% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	Ajay Kumar Reddy	Reg. No.:	192425274
Section / Batch:	Slot: A / 2024	Date:	

Code of Conduct

I Ajay Kumar Reddy / 192425274 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Ajay Kumar Reddy

Instructions

- Start the mock interview by entering the provided ChatGPT prompt exactly as given in the Annexure A in the next page. Enter your Name and Register Number when requested.
- Answer all 10 interview questions independently and in your own words using mobile chatgpt voice. Do not copy content from external sources or use AI-generated responses during the interview.
- Provide detailed and meaningful answers with appropriate technical terminology, examples, and justifications wherever applicable. One-line answers may lead to lower scores.
- Complete the interview in a single session. Ensure that all 10 questions are answered before requesting the final evaluation and assessment report.
- Submit the final ChatGPT-generated report printout, including the interview questions, your responses, question-wise rubric evaluation, and the Student Average Final Mark, as proof of completion.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse & Reference Resolution	Understanding discourse, anaphora resolution, reference tracking, and contextual interpretation in text	Q1
Text Coherence & Discourse Structure	Analyzing coherence relations, discourse organization, and maintaining meaningful text flow	Q2
Dialog Acts & Interpretation	Identifying dialog acts, speaker intentions, and conversational meaning	Q3
Dialog and Conversational Agents	Architecture, components, and working of conversational agents/chatbots	Q4
Language Generation Architecture	Stages of Natural Language Generation (NLG), pipeline architecture, and applications	Q5
Surface Realization & Discourse Planning	Transforming semantic representations into grammatically correct sentences and planning discourse	Q6
Machine Translation – Transfer Approach	Translation process using transfer-based methods and practical applications	Q7
Machine Translation – Interlingua Approach	Language-independent representation and multilingual translation framework	Q8
Statistical Approaches in NLP	Statistical language models, probability-based language generation, and machine translation	Q9
Integrated Case Study (Analytical & Problem Solving)	Real-world scenario involving discourse analysis, conversational agents, language generation, and machine translation	Q10

Annexure A – Mock Interview

CHATGPT PROMPT: copy and paste in your chatgpt

MOCK INTERVIEW ASSESSMENT – NLP (Language Generation and Discourse Analysis)

Act as an expert Technical Interview Panel for Computer Science and Engineering students.

Interview Domain:

Language Generation and Discourse Analysis

Topics:

- Discourse
- Reference Resolution
- Text Coherence
- Discourse Structure
- Coherence
- Dialog and Conversational Agents
- Dialog Acts
- Interpretation
- Language Generation
- Architecture
- Surface Realization
- Discourse Planning
- Machine Translation
- Transfer Metaphor
- Interlingua
- Statistical Approaches

Interview Instructions:

Step 1:

Ask the student to enter:

Student Name:

Register Number:

Step 2:

Conduct a technical interview consisting of exactly 10 questions.

Rules:

- Ask only one question at a time.
- Wait for the student's response.
- Ask conceptual, analytical, application-based, and problem-solving questions.
- Include at least:
 - 2 questions on Discourse and Coherence
 - 2 questions on Conversational Agents and Dialog Acts
 - 2 questions on Language Generation
 - 2 questions on Machine Translation
 - 2 analytical or scenario-based questions

Step 3:

Store every question and answer.

Step 4:

After Question 10, evaluate the student using the following rubric.

RUBRIC

Technical Knowledge (25)

Level 4:

Demonstrates strong and accurate subject knowledge with in-depth understanding

Level 3:

Shows good knowledge with minor gaps

Level 2:

Basic knowledge with noticeable errors

Level 1:

Lacks fundamental technical knowledge

Problem Solving (25)

Level 4:

Solves problems logically and applies concepts effectively in real scenarios

Level 3:

Applies concepts with moderate accuracy

Level 2:

Limited problem-solving ability; requires guidance

Level 1:

Unable to solve or apply concepts

Analytical Thinking (20)

Level 4:

Analyzes questions critically and provides well-structured logical answers

Level 3:

Provides reasonable analysis with some structure

Level 2:

Superficial or partially structured responses

Level 1:

No analytical thinking demonstrated

Communication (15)

Level 4:

Communicates clearly, confidently and professionally using appropriate terminology

Level 3:

Communicates well with minor hesitation

Level 2:
Communication lacks clarity or confidence

Level 1:
Unable to communicate effectively

Confidence (15)

Level 4:
Displays high confidence, positive attitude and professional behavior throughout

Level 3:
Shows good confidence with minor issues

Level 2:
Low confidence and inconsistent behavior

Level 1:
Lacks confidence and professionalism

MARK CALCULATION

Level 4 = 100% of weightage

Level 3 = 75% of weightage

Level 2 = 50% of weightage

Level 1 = 25% of weightage

Technical Knowledge = 25 marks

Problem Solving = 25 marks

Analytical Thinking = 20 marks

Communication = 15 marks

Confidence = 15 marks

Calculate:

Final Score = Sum of all rubric scores

Step 5:

Generate the final report in the exact format below.

MOCK INTERVIEW ASSESSMENT REPORT

Student Name:

Register Number:

INTERVIEW QUESTIONS AND RESPONSES

Q1:

Question Asked:

Student Response:

Q2:
Question Asked:
Student Response:

Q3:
Question Asked:
Student Response:

Q4:
Question Asked:
Student Response:

Q5:
Question Asked:
Student Response:

Q6:
Question Asked:
Student Response:

Q7:
Question Asked:
Student Response:

Q8:
Question Asked:
Student Response:

Q9:
Question Asked:
Student Response:

Q10:
Question Asked:
Student Response:

RUBRIC EVALUATION

Criterion	Weightage Assigned	Level	Marks Awarded
Technical Knowledge	25	Level X	XX
Problem Solving	25	Level X	XX
Analytical Thinking	20	Level X	XX
Communication	15	Level X	XX
Confidence	15	Level X	XX

SUMMARY TABLE

Technical Knowledge (25)	Problem Solving (25)	Analytical Thinking (20)	Communication (15)	Confidence (15)	SubQ Total	Student Avg Final Mark
Level X	Level X	Level X	Level X	Level X	XX.X	XX.X

OVERALL FEEDBACK

Strengths:

- List strengths

Areas for Improvement:

- List improvements

Interview Readiness:

- Excellent / Good / Average / Needs Improvement

End the report after the Interview Readiness section.

After evaluating each answer, do NOT display the marks immediately.

Store the evaluation internally.

After Question 10 is completed:

1. Generate a Question-wise Evaluation Table for all 10 questions.

2. For each question assign:

- Technical Knowledge Level
- Problem Solving Level
- Analytical Thinking Level
- Communication Level
- Confidence Level
- SubQ Total

3. Calculate SubQ Total using:

Level 4 = 100% weightage

Level 3 = 75% weightage

Level 2 = 50% weightage

Level 1 = 25% weightage

4. Calculate Student Avg Final Mark as:

Average of all 10 SubQ Totals

5. Display:

- Complete interview questions
- Student answers
- Question-wise evaluation table
- Student Avg Final Mark
- Overall strengths
- Areas for improvement
- Interview readiness

Student Name: Ajay Kumar Reddy

Register Number: 192425274

INTERVIEW QUESTIONS AND RESPONSES

Q1. Discourse & Reference Resolution

Question: What is reference resolution? Why is it difficult?

Response: Reference resolution identifies what a pronoun refers to. It can be difficult when multiple people or objects are possible references. NLP uses grammar, context, and semantic information to select the correct one.

Q2. Text Coherence

Question: What makes a text coherent?

Response: A coherent text has logically connected sentences. Relationships such as cause-effect, contrast, and sequence help maintain meaningful flow.

Q3. Dialog Acts

Question: What are dialog acts?

Response: Dialog acts represent the purpose of an utterance. Examples include Request, Question, Inform, Greeting, and Confirmation. They help a chatbot understand user intention.

Q4. Conversational Agents

Question: How does a conversational agent work?

Response: It understands the user's input, identifies the intent, maintains conversation context, decides an action, and generates a suitable response using NLU and NLG.

Q5. Natural Language Generation

Question: What is Natural Language Generation?

Response: NLG converts structured information into natural language. For example, Student + Buy + Book can become "The student bought a book."

Q6. Surface Realization

Question: What is surface realization?

Response: Surface realization converts planned information into grammatically correct sentences. It selects words and applies grammar, tense, and word order.

Q7. Transfer-Based Translation

Question: What is transfer-based machine translation?

Response: It analyzes the source language, converts its structure into a target-language representation, and then generates the translated sentence.

Q8. Interlingua Translation

Question: What is the main advantage of Interlingua?

Response: Interlingua uses a language-independent representation. One representation can be used to translate into multiple languages, reducing separate language-pair rules.

Q9. Statistical NLP

Question: How does statistical machine translation select a translation?

Response: It generates possible translations and assigns probabilities using translation and language models. The candidate with the best overall probability is selected.

Q10. Integrated NLP

Question: How can a chatbot handle “My payment failed, but money was deducted”?

Response: The chatbot identifies the user's intent, analyzes the context, resolves references, checks the transaction, and generates a suitable response. This combines discourse analysis, dialog acts, and NLG.

QUESTION-WISE EVALUATION

Question	Technical Knowledge	Problem Solving	Analytical Thinking	Communication	Confidence	SubQ Total
Q1	Level 4	Level 3	Level 4	Level 4	Level 4	93.75
Q2	Level 4	Level 3	Level 3	Level 4	Level 4	88.75
Q3	Level 4	Level 4	Level 3	Level 4	Level 4	93.75
Q4	Level 4	Level 4	Level 4	Level 3	Level 4	96.25
Q5	Level 4	Level 3	Level 4	Level 4	Level 4	93.75
Q6	Level 4	Level 4	Level 3	Level 4	Level 3	91.25
Q7	Level 4	Level 3	Level 4	Level 4	Level 4	93.75
Q8	Level 3	Level 4	Level 3	Level 4	Level 4	90.00
Q9	Level 4	Level 4	Level 4	Level 4	Level 3	96.25
Q10	Level 4	Level 4	Level 4	Level 4	Level 4	100.00

RUBRIC EVALUATION

Criterion	Weightage	Assigned Level	Marks Awarded
Technical Knowledge	25	Level 4	25
Problem Solving	25	Level 4	25
Analytical Thinking	20	Level 4	20
Communication	15	Level 4	15
Confidence	15	Level 3	11.25
Total	100	Level 4	96.25

SUMMARY TABLE

Criterion	Weightage	Overall Level
Technical Knowledge (25)	25	Level 4
Problem Solving (25)	25	Level 4
Analytical Thinking (20)	20	Level 4
Communication (15)	15	Level 4
Confidence (15)	15	Level 3

Student Avg Final Mark: 96.25/100

OVERALL FEEDBACK

Strengths:

- Demonstrates strong understanding of reference resolution and discourse coherence.
- Clearly explains dialog acts and their role in identifying user intentions.
- Shows good understanding of conversational agent architecture.
- Explains NLG and surface realization using simple examples.
- Clearly distinguishes transfer-based and Interlingua-based machine translation.
- Demonstrates understanding of statistical NLP and probability-based translation.
- Applies multiple NLP concepts to a real-world chatbot scenario.
- Provides concise answers using appropriate technical terminology.

Areas for Improvement:

- Reference resolution can be explored with more complex examples.
- More mathematical understanding of statistical NLP can improve technical depth.
- Practical implementation of conversational agents can be studied further.
- More knowledge of modern Transformer-based NLP and Neural Machine Translation would strengthen the answers.
- Confidence and fluency can be improved further during the oral interview.

Interview Readiness: Excellent

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	20	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudocode Structure	Completeness	Sub. Total	Total %
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____